United Nations

Our Common Future
Chapter 11: Peace, Security, Development, and the Environment

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1. Among the dangers facing the environment, the possibility of nuclear war, or military conflict of a lesser scale involving weapons of mass destruction, is undoubtedly the gravest. Certain aspects of the issues of peace and security bear directly upon the concept of sustainable development. Indeed, they are central to it.

2. Environmental stress is both a cause and an effect of political tension and military conflict./1 Nations have often fought to assert or resist control over raw materials, energy supplies, land, river basins, sea passages, and other key environmental resources. Such conflicts are likely to increase as these resources become scarcer and competition for them increases.

3. The environmental consequences of armed conflict would be most devastating in the case of thermo-nuclear war. But there are damaging effects too from conventional, biological, and chemical weapons, as well as from the disruption of economic production and social organization in the wake of warfare and mass migration of refugees. But even where war is prevented, and where conflict is contained, a state of 'peace' might well entail the diversion into armament production of vast resources that could, at least in part, be used to promote sustainable forms of development.

4. A number of factors affect the connection between environmental stress, poverty, and security, such as inadequate development policies, adverse trends in the international economy, inequities in multi-racial and multi-ethnic societies, and pressures of population growth. These linkages among environment, development, and conflict are complex and, in many cases, poorly understood. But a comprehensive approach to international and national security must transcend the traditional emphasis on military power and armed competition The real sources of insecurity also encompass unsustainable development, and its effects can become intertwined with traditional forms of conflict in a manner that can extend and deepen the latter.

I. Environmental Stress as a Source of Conflict

5. Environmental stress is seldom the only cause of major conflicts within or among nations. Nevertheless, they can arise from the marginalization of sectors of the population and from ensuing violence. This occurs when political processes are unable to handle the effects of environmental stress resulting, for example, from erosion and desertification. Environmental stress can thus be an important part of the web of causality associated with any conflict and can in some cases be catalytic.

6. Poverty, injustice, environmental degradation, and conflict interact in complex and potent ways. One manifestation of growing concern to the international community is the phenomenon of 'environmental refugees'/2 The immediate cause of any mass movement of refugees may appear to be political upheaval and military violence. But the underlying causes often include the deterioration of the natural resource base and its capacity to support the population.

7. Events in the Horn of Africa are a case in point. In the early 1970s, drought and famine struck the nation of Ethiopia. Yet it has been found that the hunger and human misery were caused more by years of overuse of soils in the Ethiopian highlands and the resulting severe erosion than by drought. A report commissioned by the Ethiopian Relief and Rehabilitation Commission found: 'The primary cause of the famine was not drought of unprecedented severity, but a combination of long-continued bad land use and steadily increased human and stock populations over decades'./3

8. Wars have always compelled people to leave their homes and their lands, to become refugees. Also, the wars in our time have forced large numbers of people to leave their homelands. In addition, we now have the phenomenon of environmental refugees. In 1984-85, some 10 million Africans fled their homes, accounting for two-thirds of all refugees worldwide. Their flight was not surprising in a region where 35 million suffered from famine. Many of them swarmed into cities. But many others moved across national boundaries, heightening interstate tensions. Cote d'Ivoire, Ghana, and Nigeria have been generous in welcoming refugees from the desertified Sahel. Tanzania, Zambia, and Zimbabwe have also been receiving large numbers of refugees. Yet, the Cote d'Ivoire, for instance, which depends for much of its export revenues on timber, is suffering rapid deforestation caused in part by land hunger, and one-third of landless people are immigrants. Agriculture destroys 4.5 times as much forestland in the Cote d'Ivoire as logging does./4

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9. Almost 1 million Haitian 'boat people', one-sixth of the entire populace, have fled that island nation, an exodus fuelled in large part by environmental degradation. Haiti suffers some of the world's most severe erosion, down to bedrock over large parts of some regions, so that even farmers with reasonable amounts of land cannot make a living. According to a US Agency for International Development (USAID) report, 'The social and economic effects of environmental degradation are great, and contribute to the growing outflow from rural areas. Thousands of rural Haitians leave their homes each year for Port au Prince, other Caribbean islands and the United States in search of employment and better living conditions./5 El Salvador, one of the most troubled nations of Central America, is also one of the most environmentally impoverished, with some of the worst erosion rates in the region. The fundamental causes of the present conflict are as much environmental as political, stemming from problems of resource distribution in an overcrowded land, according to a draft USAID environmental profile of El Salvador./6

> Today we cannot secure security for one state at the expense of the other. Security can only be universal, but security cannot only be political or military, it must be as well ecological, economical, and social. It must ensure the fulfilment of the aspirations of humanity as a whole.
>
> A. S. Timoshenko
> Institute of State and Law, USSR Academy of Sciences
> WCED Public Hearing
> Moscow, 11 Dec 1986

10. South Africa reveals similar problems. The inhuman policy of apartheid is at the core of the state of political conflict in Southern Africa. One of the many ways by which apartheid institutionalizes both conflict and environmental degradation is by allocating, through the 'homelands' system, 14 per cent of the nation's land to 72 per cent of the population./7 Young working-age blacks flee the overcultivated and overgrazed 'homelands' to seek work in the cities, where, on top of the squalor of overcrowded townships, they encounter extreme socio-economic inequality and racial segregation. They fight back. Repression intensifies, and the victims seek refuge over the border - whereupon the South African regime widens the conflict into neighbouring states. The entire region is becoming caught up in the ensuing violence, which could well ignite wider conflict drawing in major powers.

11. In addition to the interrelated problems of poverty, injustice, and environmental stress, competition for non-renewable raw materials, land, or energy can create tension. It was the quest for raw materials that underlay much of the competition between colonial powers and the subjugation of their holdings. Conflicts in the Middle East inevitably contain the seeds of great power intervention and global conflagration, in part because of the international interest in oil.

12. As unsustainable forms of development push individual countries up against environmental limits, major differences in environmental endowment among countries, or variations in stocks of usable land and raw materials, could precipitate and exacerbate international tension and conflict. And competition for use of the global commons, such as ocean fisheries and Antarctica, or for use of more localized common resources in fixed supply, such as rivers and coastal waters, could escalate to the level of international conflict and to threaten international peace and security.

13. Global water use doubled between 1940 and 1980, and it is expected to double again by 2000, with two thirds of the projected water use going to agriculture. Yet 80 countries, with 40 per cent of the world's population, already suffer serious water shortages./8 There will be growing competition for water for irrigation, industry, and domestic use. River water disputes have already occurred in North America (the Rio Grande), South America (the Rio de la Plata and Parana), South and Southeast Asia (the Mekong and the Ganges), Africa (the Nile), and the Middle East (the Jordan, Litani, and Orontes, as well as the Euphrates).

> How can the world of nature and the community of peoples with their national economies be harmonized? Posing the question this way suggests that the two are separate. But not so. Humanity, the human species, exists and it supported within the world of nature. And I mean that not figuratively but literally.
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> We are deep-air animals living inside an ecological system. We draw boundaries, of course, on the ecosphere for national and regional purposes. But it is all of one piece.

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When, therefore, we optimistically declare that economic development and environmental maintenance can go along hand in hand, this qualifier must immediately be added: only if the maintenance of the ecosphere is made the first priority. Economic development must be secondary, guided by strict ecological standards. These fundamental ideas are far from being universally accepted.

Stanley Rowe
Saskatchewan Environmental Society
WCED Public Hearing
Ottawa, 26-27 May 1986

14. Fisheries, whether coastal or oceanic, are fundamental to the diets of many countries. For some countries, fishing is a key economic sector, and overfishing poses immediate dangers to several national economies. In 1974 Iceland, largely dependent on its fishing industry, found itself embroiled with the United Kingdom in a 'cod war'. Similar tensions exist in the Japanese and Korean seas and on both sides of the South Atlantic. The 1986 declaration of an exclusive fishery zone around the Falkland/Malvinas Islands has further unsettled relations between Britain and Argentina. Disputes over fishing rights in the South Pacific and the search for tuna by distant-water fleets led to increased competition for diplomatic and fisheries advantages by the major powers in that region in 1986. Fisheries-related disputes may well become more frequent as nations harvest fish stocks beyond the level of sustainable yields.

15. Environmental threats to security are now beginning to emerge on a global scale. The most worrisome of these stem from the possible consequences of global warming caused by the atmospheric build-up of carbon dioxide and other gases./9 (See Chapter 7.) Any such climatic change would quite probably be unequal in its effects, disrupting agricultural systems in areas that provide a large proportion of the world's cereal harvests and perhaps triggering mass population movements in areas where hunger is already endemic. Sea levels may rise during the first half of the next century enough to radically change the boundaries between coastal nations and to change the shapes and strategic importance of international waterways - effects both likely to increase international tensions. The climatic and sea-level changes are also likely to disrupt the breeding grounds of economically important fish species. Slowing, or adapting to, global warming is becoming an essential task to reduce the risks of conflict.

II. Conflict as a Cause of Unsustainable Development

16. Arms competition and armed conflict create major obstacles to sustainable development. They make huge claims on scarce material resources. They pre-empt human resources and wealth that could be used to combat the collapse of environmental support systems, the poverty, and the underdevelopment that in combination contribute so much to contemporary political insecurity. They may stimulate an ethos that is antagonistic towards cooperation among nations whose ecological and economic interdependence requires them to overcome national or ideological antipathies.

17. The existence of nuclear weapons and the destructive potential inherent in the velocity and intensity of modern conventional warfare have given rise to a new understanding of the requirements for security among nations. In the nuclear age nations can no longer obtain security at each other's expense. They must seek security through cooperation, agreements, and mutual restraint; they must seek common security./10 Hence interdependence, which is so fundamental in the realm of environment and economics, is a fact also in the sphere of arms competition and military security. Interdependence has become a compelling fact, forcing nations to reconcile their approach to 'security'.

1. Nuclear War - Threat to Civilization

18. The likely consequences of nuclear war make other threats, to the environment pale into insignificance. Nuclear weapons represent a qualitatively new step in the development of warfare. One thermo-nuclear bomb can have an explosive power greater than that of all the explosives Used in wars since the invention of gunpowder. In addition to the destructive effects of blast and heat, immensely magnified by these weapons, they introduce a new lethal agent - ionizing radiation - that extends lethal effects over both space and tine.

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19. In recent years, scientists have in addition called our attention to the prospect of 'nuclear winter'. It has been most authoritatively explored by some 300 scientists from the United States, the USSR, and more than 30 other countries - working on a collaborative basis in some cases across ideological divides./11

> All youth organizations believe that environmental issues stand high on the priority list of global problems. However, their solution depends on the preservation of peace on our planet. The quest of solutions to ecological problems is impossible without the curbing of the arms race, for the arms race absorbs tremendous intellectual and material resources of mankind. The solution of ecological problems also depends on the way of life of young people and their value orientation.
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> Dr. I.I. Russin
> Moscow State University
> WCED Public Hearing
> Moscow, 8 Dec 1986

20. The theory contends that the smoke and dust ejected into the atmosphere by a nuclear war could absorb enough solar radiation to remain aloft for some time, preventing sunlight from reaching the surface of the earth, causing a widespread and prolonged cooling of land areas. There would be severe repercussions for plant life generally and for agriculture in particular, disrupting the production of food to sustain survivors of the war. Great uncertainties remain about the scale and linkages determining environmental effects, but large-scale environmental perturbations are considered probable. A nuclear war cannot be won, and must never be fought. In the aftermath, there would be no difference between so called victor and vanquished. The nuclear-weapon states must spare no effort to conclude a verifiable agreement on banning all nuclear weapon tests.

21. The findings on nuclear winter are vitally important too for non-aligned nations, predominantly in the South, which are not parties to the East West conflict. They cannot expect to avoid the potentially disastrous environmental consequences of nuclear war in the northern hemisphere. The aftermath of such a war would envelop the world. There is a danger that nuclear weapons will spread to more and more countries and be used in what begins as a limited regional conflict. Beyond the five recognized nuclear-weapon states, at least six others have a widely acknowledged potential nuclear weapons capability; a dozen others are not far behind. The nuclear-weapon states cannot expect the non-nuclear-weapon states to abstain from exercising the nuclear option in the absence of real progress on the road to nuclear disarmament. It is imperative, therefore, that the probable consequences of nuclear war be recognized universally and that all states become involved in efforts to prevent the proliferation - and above all the use of nuclear weapons.

2. Other Weapons of Mass Destruction

22. Other forms of war and other weapons of mass destruction have large scale effects or both human societies and the human environment. Biological warfare could release new agents of disease that would prove difficult to control. Recent advances in biotechnology multiply the potentially lethal applications of such weapons. Likewise, the deliberate manipulation of the environment (for example, through artificial earthquakes and floods) would have consequences far beyond the borders of those involved in a conflict, were they ever used. Chemical agents can seriously damage the environment, as demonstrated by the defoliants used in South-east Asia. The dangerous and environmentally unpredictable consequences of biological and chemical weapons have led to international agreements banning their use./12 But there is need for further efforts to strengthen the regimes to which these agreements contribute. In particular, the Geneva protocol prohibiting the use of chemical weapons should be supplemented by agreements prohibiting the production and stockpiling of such weapons.

23. Military applications of new technologies now threaten lo make outer space a focus of international competition and conflict. (See Chapter 10.) Most countries in the international community see space as a global commons that should benefit humanity as a whole and be preserved from military competition - a feeling reflected in the 1967 Outer Space Treaty, under which nations agreed not to deploy weapons of mass destruction there. Governments should now agree on measures to prevent an arms race in space and stop it on Earth. Failing such agreement, the arms race could expand, with dire consequences for humanity.

3. The Costs of the 'Arms Culture'

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24. The absence of war is not peace; nor does it necessarily provide the conditions for sustainable development. Competitive arms races breed insecurity among nations through spirals of reciprocal fears. Nations need to muster resources to combat environmental degradation and mass poverty. By misdirecting scarce resources, arms races contribute further to insecurity.

25. The coexistence of substantial military spending with unmet human needs has long evoked concern. President Eisenhower, for example, observed at the end his term in office that 'every gun that is made, every warship launched, every rocket fired represents, in the final analysis, a theft from those who hunger and are not fed, who are cold and are not clothed'./13

26. Global military spending in 1985 was well in excess of $900 billion./14 This was more than the total income of the poorest half of humanity. It represented the equivalent of almost $1,000 for every one of the world's 1 billion poorest. Put another way, military spending surpassed the combined gross national products of China, India, and the African countries south of the Sahara. Moreover, global military spending has risen not only absolutely but proportionately - from an estimated 4.7 per cent of world output in 1960 to over 6 per cent - representing an increase of about 150 per cent in real (constant price) terms. Three-quarters of current expenditure is in the industrial world./15

27. The true cost of the arms race is the loss of what could have been produced instead with scarce capital, labour skills, and raw materials. The plants that manufacture weapons, the transport of those weapons, and the mining of minerals for their production all place enormous demands on energy and mineral resources and are a major contributor to pollution and environmental deterioration.

28. The distorting effects of the 'arms culture' are most striking in the deployment of scientific personnel. Half a million scientists are employed on weapons research world-wide, and they account to: around half of all research and development expenditure./16 This exceeds the total combined spending on developing technologies for new energy sources, improving human health, raising agricultural productivity, and controlling pollution. Military research and development - $70-80 billion world-wide in 1984 - is growing at twice the rate of military spending as a whole./17 At the same time, there is a paucity of resources available for monitoring global climatic change, for surveying the ecosystems of disappearing rain forests and spreading deserts, and for developing agricultural technologies appropriate to rainfed, tropical agriculture.

29. Nations are seeking a new era of economic growth. The level of spending on arms diminishes the prospects for such an era - especially one that emphasizes the more efficient use of raw materials, energy, and skilled human resources. It also has a bearing, albeit indirect, on the willingness of rich countries to provide development assistance to developing countries. Clearly, there is no simple correspondence between reduced defence spending and increased aid. There are other reasons aside from domestic resource constraints for a reluctance to expand aid, and nations cannot wait for disarmament before devoting more resources to ensuring sustainable development. Nonetheless, increased defence spending puts pressure on other budgetary items, and aid is an easy target, despite being a relatively small outlay for most donor countries./18

30. Although redeployment is clearly possible, resources currently employed in military applications cannot be redeployed quickly or easily elsewhere in other sectors or other countries. There are technical problems in achieving such a transformation, not least the contribution made by military spending to jobs in economies with high unemployment. And beyond the technical problems are questions of political will. Nonetheless, some countries - China, Argentina, and Peru, for example - have recently shown that it is both technically and politically possible to make substantial shifts from military to civilian spending within a short time./19

4. World Armaments and the Growth of the 'Arms Culture'

31. Traditionally, nations have adhered to an 'arms culture'. They find themselves locked into arms competitions fuelled among other things by powerful vested interests in the 'military-industrial complex' as well as in the armed forces themselves. Industrial nations account for most of the military expenditures and the production and transfer of arms in international society. However, the influence of this 'arms culture' is not confined to these nations. It is present also in the developing world, fostered both by the desire of many governments to seek security through acquisition of arms and by a burgeoning world arms trade.

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I have here listened to people speaking about financial crises, famine, pollution, and social injustice at various levels. As an ecologist, I cannot see any of these questions without linking them to the armaments question and to the nuclear issue.

Poverty generates tensions and conflicts, urban and rural violence. The indigenous people are still awaiting solutions for their problems. All this depends on money and nevertheless we are spending money on our nuclear programmes. They say that this has peaceful objectives. This is not true because precious money is being spent on this.

The greatest crime: the death of hope, the death of all of the rights we all have, especially that of the young of believing in a future, the hope for a normal life, a difficult life but something that appears as a challenge to live it the best we can. We have a right to this chance.

Cacilda Lanuza
Brazilian Ecological Movement
WCED Public Hearing
Sao Paulo, 28- 29 Oct 1985

32. Since the early 1960s, military spending in developing countries as a whole has increased fivefold. Their share of total spending increased from under one-tenth to almost a quarter of a far larger total./20 Some developing countries, such as the Republic of Korea, have achieved a high level of development in spite of military spending. But systematic analysis suggests that increases in military spending have had negative effects on economic performance./21

33. Moreover, defence expenditure is one of the most import-intensive of activities, usually creating a large secondary demand for imported spares, ammunition, servicing, training, and fuel. It has been estimated that 20 per cent of the external debt acquired by non-oil developing countries in the decade to 1982 could be attributed to arms imports./22 And high levels of arms spending motivated by a variety of reasons have undoubtedly contributed to the severity of the crises of development in Africa, where military spending rose, in real terms, by 7.8 per cent per annum between 1971 and 1982, and arms imports rose by 18.5 per cent./23 it should be noted in this connection that in the case of the Frontline States they have been compelled to expand their armed forces because of the threat from South Africa.

34. The development of an 'arms culture' in many developing countries presents particular dangers in the context of environmental and poverty-induced stresses. There are already numerous simmering disputes in the Third World - over 40 unresolved - many arising from boundaries defined in colonial times.

35. Sophisticated weapons can help convert the potential into actual conflict. According to the UN Group of Governmental Experts on the Relationship Between Disarmament and Development:

> There can no longer be the slightest doubt that resource scarcities and ecological stresses constitute real and imminent threats to the future well-being of all people and nations. These challenges are fundamentally non-military and it is imperative that they be addressed accordingly. If this is not recognized, ... there is a grave risk that the situation will deteriorate to the point of crisis where, even with low probability of success, the use of force could be seen as a way to produce results quickly enough. This is far from being a remote possibility. In recent years, there has been a marked tendency in international relations to use or to threaten to use military force in response to non-military challenges to security./24

36. The situation in many developing countries presents particular dangers in the context of environmental and poverty-induced stresses. Large-scale movements of refugees, competition for scarce water and fertile lands, deposits of oil and raw materials, ill-defined boundaries, and so on all add to tensions and increase possibilities for conflict. The importation of armaments by developing countries has increased also because of these real or potential conflicts. It is sometimes encouraged by the arms manufacturers because of the important profits that can themselves sustain the manufacture of arms in the exporting countries. The export of arms have been evaluated at more than $35 billion annually. The arms trade is estimated to have absorbed over $300 billion over the last two decades, three-quarters in the form of sales to developing countries./25

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III. Towards Security and Sustainable Development

1. Principles

37. The first step in creating a more satisfactory basis for managing the interrelationships between security and sustainable development is to broaden our vision. Conflicts may arise not only because of political and military threats to national sovereignty; they may derive also from environmental degradation and the pre-emption of development options.

38. There are, of course, no military solutions to 'environmental insecurity'. And modern warfare can itself create major internationally shared environmental hazards. Furthermore, the idea of national sovereignty, has been fundamentally modified by the fact of interdependence in the realm of economics, environment, and security. The global commons cannot be managed from any national centre: The nation state is insufficient to deal with threats to shared ecosystems. Threats to environmental security can only be dealt with by joint management and multilateral procedures and mechanisms.

> Environment must also be an approach to development. Environment is a social justice issue and environment even is a peace and security issue. The barriers to achieving sustainable development are great, as might be expected in a major historical transformation, but they are far from insurmountable.
>
> We approach the millennium in a world in which global interdependence is the central reality, but where absolute poverty and environmental degradation cloud our vision of a common future, and where a geopolitical climate dominated by nuclear terrorism and increasing militarization saps the idealism of the young and the will to dream in us all.
>
> Ralph Torrie
> On Behalf of Canadian Environment, Development and Peace Organizations
> WCED Public Hearing
> Ottawa, 26-27 May 1986

2. Cooperative Management

39. Already, environmental stresses are encouraging cooperation among nations, giving some indication of ways to proceed. Antarctica is subject to a far-reaching agreement that provides a collective approach to management. (See Chapter 10.) There are now various institutional systems, often of complex and advanced form, to foster bilateral and regional cooperation for marine fisheries in order to regulate maximum sustainable yields and the distribution of catches. One of the main threats to the oceans the dumping of highly toxic wastes has so far been managed by the London Dumping Convention. As for international water bodies, impressive progress has been made by the bilateral U.S.-Canadian Commission for the Great Lakes. The Mediterranean Convention, only one of the many such treaties concluded within the context of the UNEP Regional Seas Programme, brings together coastal nations in an arrangement to monitor and combat pollution at sea.

40. Some of the most challenging problems require cooperation among nations enjoying different systems of government, or even subject to antagonistic relations. The 1986 Chernobyl reactor accident in the Soviet Union has resulted in two agreements covering international cooperation in cases of such accidents. In the future, the nation concerned will immediately alert neighbouring states; they, in turn, will offer assistance at cost and free of liability./26 The 1979 Convention on Transboundary Pollution has provided a framework for monitoring and assessing damage from pollutants causing acid rain in Europe./27

41. Cooperation on environmental issues among developing countries has often been made difficult by poor communications. Nonetheless, many now participate in UNEP's Regional Seas Programme. The nations of the Sahel have formed a regional organization to deal with desertification, and there ie emerging a body of successful case histories with respect to river basin development: Witness the joint management programmes in Africa for the Senegal River Basin.

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Box 11-1

Spending on Military Versus Environmental Security

The world spent well over $900 billion on military purposes in 1985, more than $2.5 billion a day. The real cost is what the same resources might otherwise be used for:

- An Action Plan for Tropical Forests would cost $1.3 billion a year over the course of five years. This annual sum is the equivalent of half a day of military expenditure worldwide.

- Implementing the UN Action Plan for Desertification would cost $4.5 billion a year during the last two decades of this century - the equivalent of less than two days of military spending.

- One of the greatest environmental hazards in the Third World is lack of clean water for household use, contributing to 80 per cent of disease. The UN Water and Sanitation Decade, although given only a small fraction of support needed, would have cost $30 billion a year during the 1980s. This is the approximate equivalent of 10 days of military spending.

- To supply contraceptive materials to all women already motivated to use family planning would cost an additional $1 billion per year on top of the $2 billion spent today. This additional $1 billion is the equivalent of 10 hours of military spending.

Sources: International Task Force, Tropical Forests: A Call for Action (Washington, DC: World Resources Institute, 1965); Dr M.K. Tolba, 'Desertification and the Economics of Survival', UNEP Information 86/2. 25 March 1986; A. Agarwal et al., Water, Sanitation and Health for All? (London: IIED/Earthscan, 1981); World Bank, World Development Report, 1984 (New York: Oxford University Press, 1984).

3. The Importance of Early Warning

42. Since it is often uncertainty and insecurity that prompts international conflict, it is of the utmost importance that governments become aware of imminent environmental stress before the damage actually threatens core national interests. Governments are usually not well equipped with this kind of foresight.

43. It would be highly desirable if the appropriate international organizations, including appropriate UN bodies and regional organizations, were to pool their resources and draw on the most sophisticated surveillance technology available - to establish a reliable early wanting system for environmental risks and conflict. (See Chapter 12.) such a system would monitor indicators of risks and potential disputes, such as soil erosion, growth in regional migration, and uses of commons that are approaching the thresholds of sustainability. The organizations would also offer their services for helping the respective countries to establish principles and institutions for joint management.

4. Disarmament and Security

44. Action to reduce environmental threats to security requires a redefinition of priorities, nationally and globally. Such a redefinition could evolve through the widespread acceptance of broader forms of security assessment and embrace military, political, environmental, and other sources of conflict.

45. A broader approach to security assessment would no doubt find many cases in which national, regional, and global security could be enhanced through expenditures quite small in relation to the levels of military spending. Four of the most urgent global environmental requirements - relating to tropical forests, water, desertification, and population - could be funded with the equivalent of less than one month's global military spending. (See Box 11-1.) It is difficult to shift budgetary resources, but individual governments have already shown that transformation is possible, given political will. In some of the countries most seriously affected by environmental stress and poverty, the sums required to alleviate these conditions are small in relation to what is now spent on disaster relief, let alone military activities./28 However, these sums must be spent quickly, before deteriorating conditions require much larger expenditures.

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46. But in terms of the aggregate resources involved in arms spending and the potential throat to the environment from war, the greatest need is to improve relations among those major powers capable of deploying weapons of mass destruction. This is needed to achieve agreement on tighter control over the proliferation and testing of various types of weapons of mass destruction nuclear and non-nuclear including those that have environmental implications./29

47. A substantial number of agreements already show the potential for negotiated, multilateral solutions. President Reagan and General Secretary Gorbachev made substantial progress towards strategic arms agreement, which must be carried forward to reverse the alarming trends of several decades. Apparently, the two major powers came close to agreeing on intermediate range systems in Europe, to be followed by agreements banning forward deployment of shorter range systems. It would alleviate significantly the pressures exercised by nuclear weapons on the security order in Europe. In addition, they are moving towards, a 50 per cent reduction agreement on strategic systems, followed by total elimination agreements. They also need to agree on effective measures to prevent an arms race in space. Successful negotiations would contribute significantly to stemming the spread of nuclear weapons as the major nuclear-weapon states would deliver on their promise to build down their nuclear arsenals. Such progress is consistent with the basic needs of our times and the right of humanity to have the spectre of nuclear destruction removed from the face of the Earth.

48. Nations must turn away from the destructive logic of an 'arms culture' and focus instead or their common future. The level of armaments and the destruction they could bring about bear no relation to the political conflict that triggered the arms competition in the first place. Nations must not become prisoners of their own arms race. They must face the common danger inherent in the weapon of the nuclear age. They must face the common challenge of providing for sustainable development and act in concert to remove the growing environmental sources of conflict.

Footnotes

1/ For some preliminary analyses along these lines, see L. Timberlake and J. Tinker, 'Environment and Conflict: Links Between Ecological Decay, Environmental Bankruptcy and Political and Military Instability, Earthscan Briefing Document. Earthscan, London, 1964; N. Myers, 'The Environmental Dimension to Security Issues', The Environmentalist, Winter 1986; R.H. Ullman, 'Redefining Security', International Security, Summer 1983; and A.H, Westing (ed.), Global Resources and International Conflict (Oxford: Oxford University Press, 1986).

2/ E. El-Hinnawi, Environmental Refugees (Nairobi: UNEP, 1985).

3/ Relief and Rehabilitation Commission, 'Drought and Rehabilitation in Wollo and Tigrai', Addis Ababa, 1975.

4/ L. Timberlake, Africa in Crisis (London: International Institute for Environment and Development/Earthscan, 1985).

5/ Project Paper for Haiti Agroforestry Outreach Project (Project 521-0122), U.S. Agency for International Development, Washington, DC, 1981.

6/ National Park Service/U.S. Man and the Biosphere Secretariat, 'Draft Environmental Profile of El Salvador', Bureau of Science and Technology, U.S. Agency for International Development, Washington, DC, April 1982. See also T.P. Anderson, The War of the Dispossessed: Honduras and El Salvador 1969 (Lincoln, Neb.: University of Nebraska Press, 1981); W.H. Durham, Scarcity and Survival in Central America: Ecological Origins of the Soccer War (Stanford, Calif.: Stanford University Press, 1979).

7/ D. Smith, 'Update: Apartheid in South Africa', Queen Mary College, London, 1984.

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10/ Independent Commission on Disarmament and Security Issues under the Chairmanship of Olof Palme, Common Security (London: Pan Books, 1982).

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12/ Outright banning of particularly lethal weapons has its origin in the St. Petersburg Declaration banning the use of 'dum-dum bullets' and the Hague war rules outlining the use of shaped charges (1899). Also relevant are the Geneva Protocol banning the military use of chemical and bacteriological weapons (1925); the Convention on the Prohibition of the Development, Production, and Stockpiling of Bacteriological and Toxin Weapons (1975); and the Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques (1978).

13/ The Eisenhower quote is taken from his final, valedictory address (Speech to the American Society of Newspaper Editors Washington, DC, April, 1953) which also includes the more famous reference to the 'military-industrial complex'.

14/ Estimates from R.L. Sivard, World Military and Social Expenditures (Washington, DC: World Priorities, Inc., 1986). More details in M. Brzoska et al., 'World Military Expenditure and Arms Production', SIPRI Yearbook, op. cit. The figure of total military spending is necessarily approximate because of the enormous problems of aggregating spending in different - and often non-convertible - currencies and from countries with different statistical conventions. According to Sivard, total military spending in 1983 was $728 billion. On the basis of trends and preliminary data, a figure of at least $900 billion and possibly $1,000 billion in current prices and exchange rates seems appropriate for 1986.

15/ Sivard, 1986 edition, op. cit., SIPRI Yearbook, op. cit.

16/ Sivard, 1986 edition, op. cit., SIPRI Yearbook, op. cit.

17/ M. Ackland-Hood, 'Military Research and Development Expenditure', SIPRI Yearbook, op. cit.

18/ According to calculations based on OECD Development Assistance Committee data, which are not universally accepted, together with Sivard, total non-military development aid measured in net concessional flows from industrial to developing countries represents roughly 5 per cent of the amount spent by all industrial countries on armaments. For the United States, foreign aid accounts for 4 per cent of armaments spending, and for the USSR, 1.5 per cent. In Austria, Denmark, the Netherlands, Norway, and Sweden, by contrast, the proportion is close to 30 per cent, and it is over 10 per cent for Australia, Belgium, Canada, France, FRG, and Switzerland,

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19/ According to L.H. Brown et al., in State of the World 1986 (London: W.W. Norton, 1986), China in 1972 spent 14 per cent of its gross national product (GNP) on military purposes, one of the highest levels in the world. Since 1970 (except for 1979), the government has systematically reduced this until by 1985 it amounted to only 7.5 per cent In mid 1985 the government announced it would cut the armed forces to 3.2 million, a drop of 24 per cent. In Argentina, by 1984 new President Raul Alfonsin had cut arms outlays to half their peak level of 1980 (nearly 4 per cent of GNP) by reordering priorities and shifting resources to social programmes. Peruvian President Alan Garcia Perez, on taking office in mid-1985, announced he would reduce military outlays, which then totalled 5 per cent of GNP, or one-quarter of the federal budget. First he cancelled half the order for 26 French Mirage fighter planes.

20/ Over 1960-81, Third World military expenditures grew by some 7 per cent per year, as compared with 3.7 per cent in the industrial world. In 1960, Third World military expenditures accounted for less than one-tenth of the global total 1981 for more than one fifth of a far larger total. R.L. Sivard, World Military and Social Expenditures (Washington, DC: World Priorities, Inc., 198b).

21/ L. Taylor, 'Military Economics in the Third World', prepared for The Independent Commission on Disarmament and Security Issues, 1981.

22/ R. Tullberg, 'Military Related Debt in Non-Oil Developing Countries', SIPRI Yearbook, op. cit.

23/ R. Luckham, 'Militarization in Africa', SIPRI Yearbook, op. cit.

24/ I. Thorsson et al., Relationship Between Disarmament and Development, Disarmament Study Review No. 5 (A/36/536) (New York: UN Department of Political and Security Council Affairs, 1982).

25/ Arms Export from L.R. Brown et al., op. cit, based on U.S. Arms Control and Disarmament Agency; estimate of cumulative spending on the arine trade in Sivard, 1985 edition, op. cit.

26/ 'Negotiations on Agreement Concerning Nuclear Safety Reach Consensus', press release (PR8-86/17), IAEA, 15 August 1986.

27/ 'Convention on Long-Range Transboundary Air Pollution' concluded 13 November 1979 and entered into force 16 March 1983, summarized in M.J. Bowman and D.J. Harris (eds.), Multilateral Treaties: Index and Current Status (London: Butterworths, 1984).

28/ The amount that the United Nations has recently budgeted for Ethiopia to cater for anti-erosion, reforestation, and related measures under its Anti-Desertification Plan suggests that no more than $50 million a year would have been required to counter much of the highlands' problem if the investment had been undertaken in due time. By contrast, the amount required to counter Ethiopia's famine during 1985 amounted to $500 million for relief measures alone. Between 1976 and 1980 Ethiopia spent an average of $225 million a year on military activities.

29/ Among international treaties specifically designed to protect the global commons from militarization are the Antarctic Treaty (1959); the Moscow Treaty Banning Nuclear Weapons Tests in the Atmosphere, in Outer Space and Under Water (1963); the Outer Space Treaty (1967); the Treaty of Tlatelolco (1967); the Treaty on the Non Proliferation of Nuclear Weapons (1968); and the Sea Bed Treaty (1971).

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